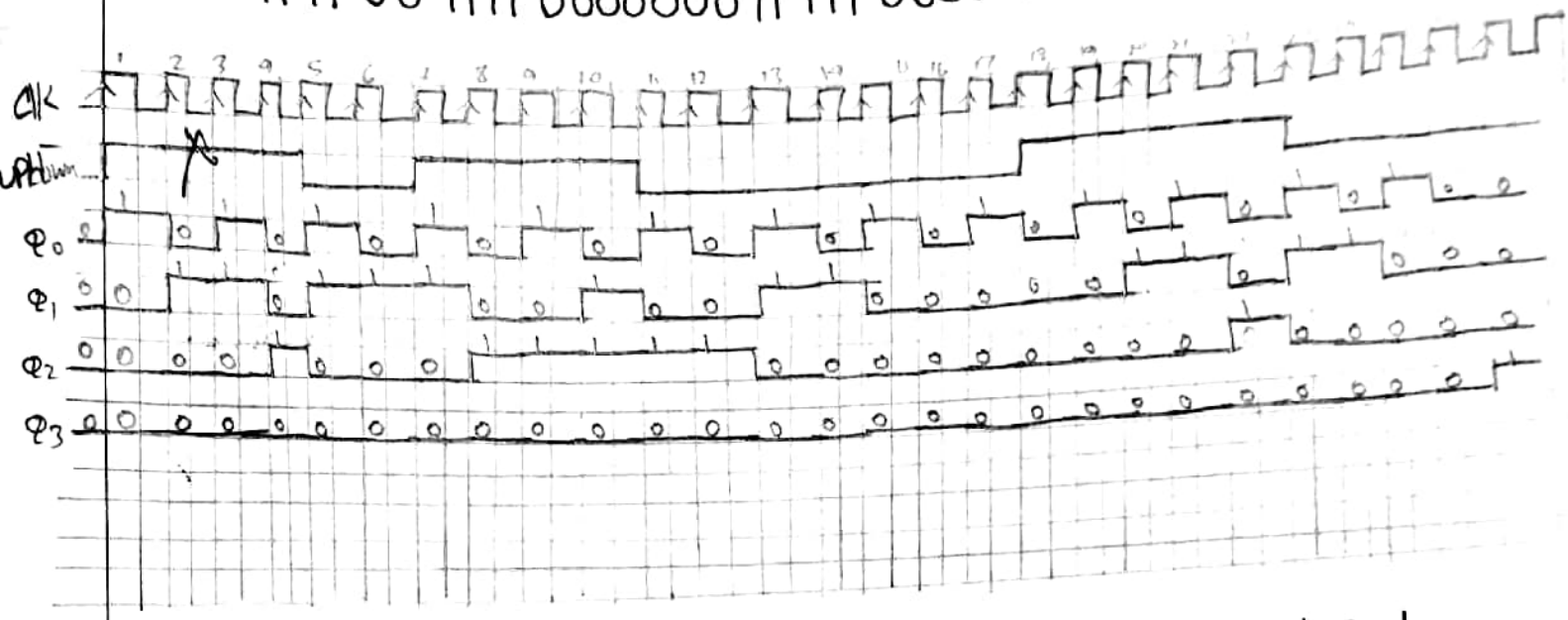
Timing diagram for given up/$\overline{\text{down}}$ input

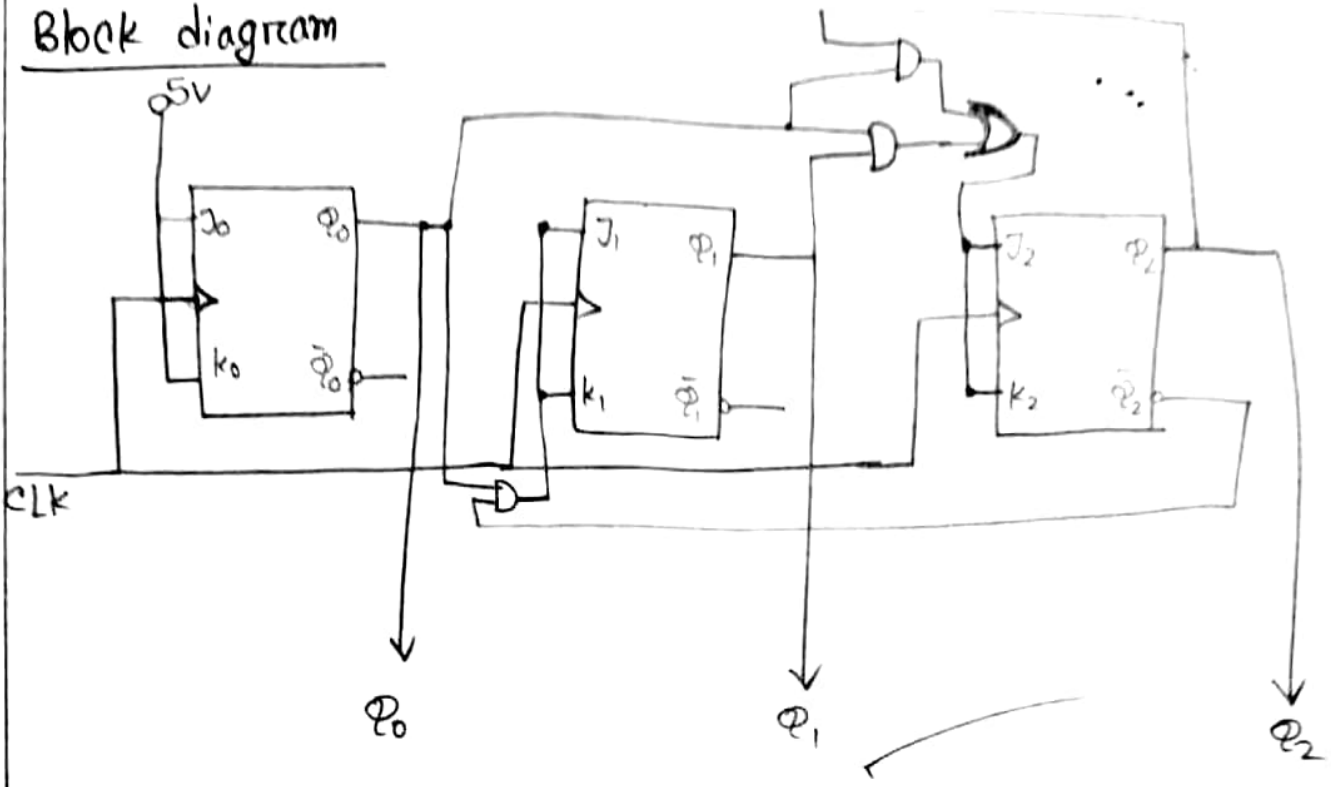
11 11 00 1111 0000000 11 111 0000 0



clk	Q_3	Q_2	Q_1	Q_0	
	0	0	0	0	
1	0	0	0	1	up
2	0	0	1	0	
3	0	0	1	1	
4	0	1	0	0	
5	0	0	1	1	down
6	0	0	1	0	
7	0	0	1	1	up
8	0	1	0	0	
9	0	1	0	1	
10	0	1	1	0	
11	0	1	1	1	down
12	0	1	0	0	
13	0	1	0	1	
14	0	0	1	0	
15	0	0	1	1	
16	0	0	0	0	
17	0	0	0	1	up
18	0	0	0	0	
19	0	0	0	1	
20	0	0	1	0	
21	0	0	1	1	
22	0	1	0	0	

CLK	Q_3	Q_2	Q_1	Q_0	
23	0	0	1	1	
24	0	0	1	0	
25	0	0	0	1	
26	0	0	0	0	
27	1	0	0	0	

Block diagram



9

Ans to Q-No-2

MOD-9 up/down synchronous counter

Input expression:

$J_0 = K_0 = (\bar{Q}_3 \cdot up) + (\overline{down} \cdot \bar{Q}_3\bar{Q}_2\bar{Q}_1\bar{Q}_0)$

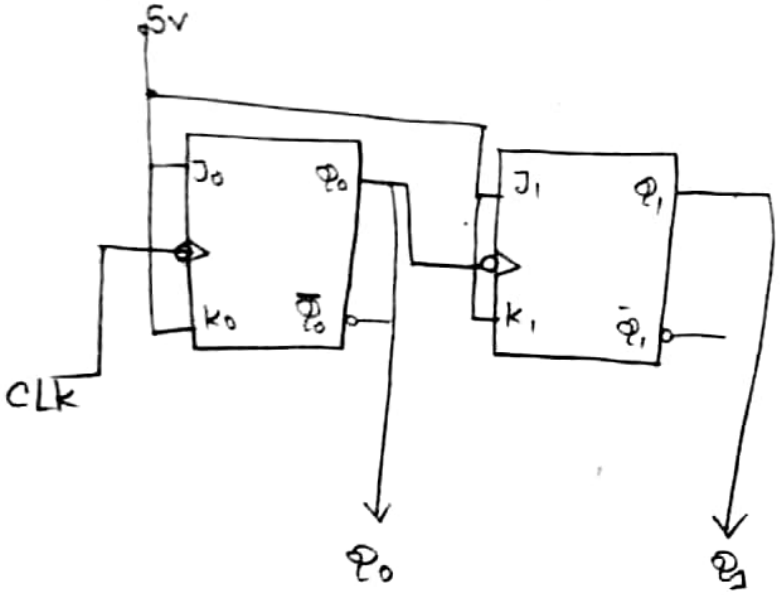
$J_1 = K_1 = (Q_0 \cdot up) +$

Truth Table

CLK	Q_3	Q_2	Q_1	Q_0
	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	0	0	0	0

asynchronous counter:

2 bit asynchronous counter



MOD-6 synchronous counter:

Input expression:

$J_0 = k_0 = 1$

$J_1 = k_1 = \bar{Q}_2 Q_0$

$J_2 = k_2 = Q_1 Q_0 + Q_2 Q_0$

Truth Table

CLK	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	0	0	0